

MTH 310/520 Graph Theory: Quiz 2

Solutions

Instructor: Manuj Mukherjee

TAs: Ashwin Singh & Mohammad Umar

Total: 10 points

1) $G = (V, E)$ is called elementary iff $\forall e \in E \exists$ a perfect matching M for the graph G with $e \in M$ and G is connected.

Suppose G is a bipartite graph with bipartition (U, W) . Then prove the following are equivalent:

- G is elementary
- G has exactly two minimum vertex covers, namely U and W ;
- $|U| = |W|$ and for every non-empty proper subset X of U , $|N(X)| > |X|$;
- $G = K_2$, or $|V| \geq 4$ and for any $u \in U, w \in W$, $G - u - w$ has a perfect matching;
- G is connected and every edge of G is allowed. [Not needed]

[10 points]

Solution:

- $(a \Rightarrow b)$ Let's look at an arbitrary K separate from U and W . We define

$$K_U = K \cap U \text{ and } K_W = K \cap W$$

since K is different from both U and W , both K_U and K_W are non-empty proper subsets of U and W , respectively. Now let's look at an edge $e = \{u, w\}$ in $G[K]$. There exists a perfect matching for M including e , since G is elementary. This perfect matching must match $U - K_U$ to K_W and $W - K_W$ to K_U , as otherwise there is an edge from $U - K_U$ to $W - K_U$ which isn't covered by the cover K . Observe that $u \in K_U$ is mapped to $w \in K_W$, hence

$$|U - K_U| \leq |K_W| - 1 < |K_W| \text{ and } |U - K_W| \leq |K_U| - 1 < |K_U|$$

Now $|U| = |U - K_U| + |K_U| < |K_W| + |K_U| = |K|$, which gives us $|U| < |K|$, this proves all other covers are not minimum.

- $(b \Rightarrow c)$ That $|U| = |W|$ is immediate since U and W are both minimum vertex covers of the same graph. Suppose, for the sake of contradiction, there is a non-empty proper subset $X \subset U$ such that $|N(X)| \leq |X|$. Consider the set $(U - X) \cup N(X)$. This set is a valid vertex cover for G because any edge not covered by $U - X$ will be covered via $N(X)$. The size of this vertex cover is bounded by:

$$|(U - X) \cup N(X)| \leq |U - X| + |N(X)| \leq |U - X| + |X| = |U|$$

As U is the minimum vertex cover $|U| = |(U - X) \cup N(X)|$

As $X \neq \emptyset$, $U \neq (U - X) \cup N(X) \neq W$ which is a contradiction to b). Hence $|N(X)| > |X|$ for all $X \subset U$

- $(c \Rightarrow d)$ Given c holds, If $|V|=2$ then d vacuously holds for K_2 . Now for $|V| \geq 4$ let $u \in U$ and $w \in W$, we show that there exists a perfect matching for $G - u - w$. Let's assume otherwise from Hall's Marriage Theorem, we can say $\exists X \subset U - u$ such that $|N_H(X)| < |X|$, but if this is true then $|N_G(X)| \leq |N_H(X)| + 1 \leq |X|$ which is a contradiction to c hence, no such X exist and hence $G - u - w$ has a perfect matching.
- $(d \Rightarrow a)$ if $G = K_2$ then we are done. For $|V| \geq 4$, suppose G is not connected then G_1 be a component of G such that $|V(G_1) \cap U| \leq |V(G_1) \cap W|$ (If no such component exists then $|U| > |W|$ which is not true). Let $u \in V(G_1) \cap U$ and $w \in W - V(G_1)$, then $G - u - w$ has no perfect matching which is a contradiction and hence G is connected. From d for every edge $e = \{u, w\}$, $G - u - w$ has a perfect matching M_H , $M_H \cup \{e\}$ is a perfect matching for G containing e . Hence $\forall e \in E \exists$ a perfect matching M for the graph G with $e \in M$ and G is connected implying G is elementary.